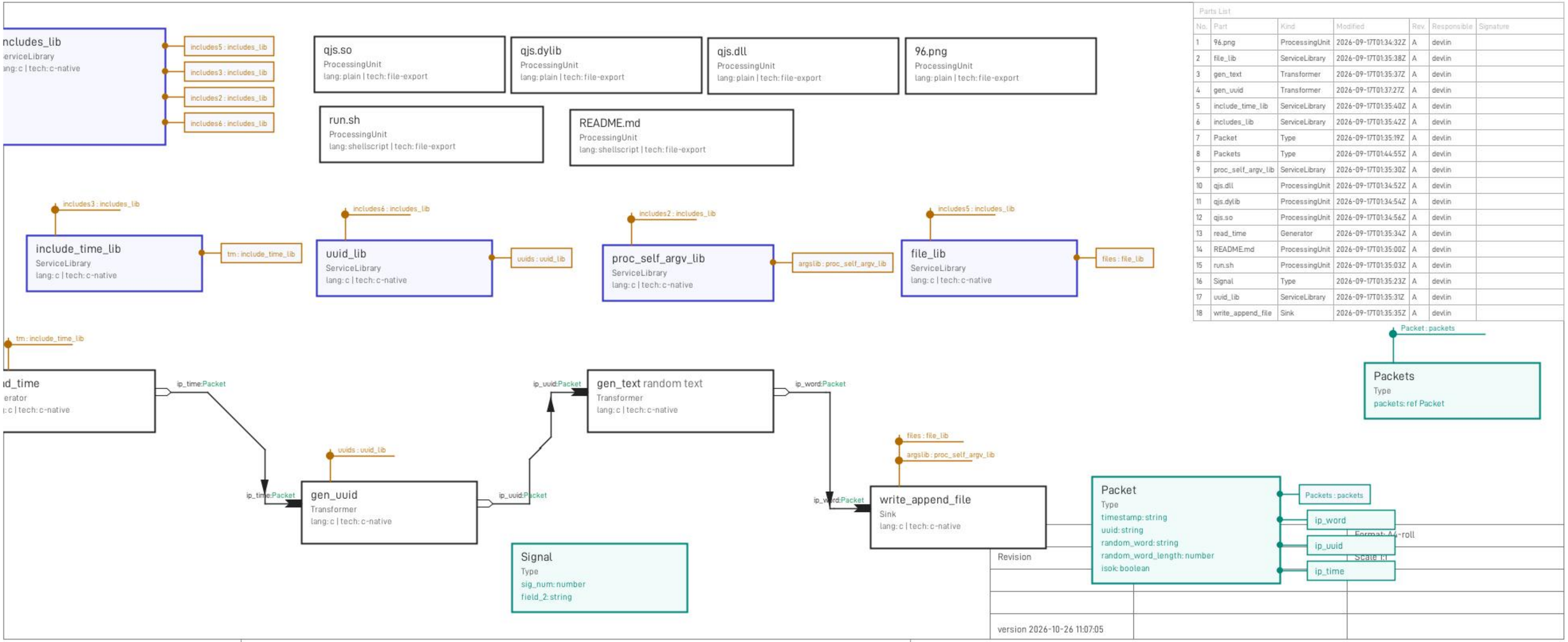


Parts List

No.	Part	Kind	Modified	Rev.	Responsible	Signature
1	96.png	ProcessingUnit	2026-09-17T01:34:32Z	A	devlin	
2	file_lib	ServiceLibrary	2026-09-17T01:35:38Z	A	devlin	
3	gen_text	Transformer	2026-09-17T01:35:37Z	A	devlin	
4	gen_uuid	Transformer	2026-09-17T01:37:27Z	A	devlin	
5	include_time_lib	ServiceLibrary	2026-09-17T01:35:40Z	A	devlin	
6	includes_lib	ServiceLibrary	2026-09-17T01:35:42Z	A	devlin	
7	Packet	Type	2026-09-17T01:35:19Z	A	devlin	
8	Packets	Type	2026-09-17T01:44:55Z	A	devlin	
9	proc_self_argv_lib	ServiceLibrary	2026-09-17T01:35:30Z	A	devlin	
10	qjs.dll	ProcessingUnit	2026-09-17T01:34:52Z	A	devlin	
11	qjs.dylib	ProcessingUnit	2026-09-17T01:34:54Z	A	devlin	
12	qjs.so	ProcessingUnit	2026-09-17T01:34:56Z	A	devlin	
13	read_time	Generator	2026-09-17T01:35:34Z	A	devlin	
14	README.md	ProcessingUnit	2026-09-17T01:35:00Z	A	devlin	
15	run.sh	ProcessingUnit	2026-09-17T01:35:03Z	A	devlin	
16	Signal	Type	2026-09-17T01:35:23Z	A	devlin	
17	uuid_lib	ServiceLibrary	2026-09-17T01:35:31Z	A	devlin	
18	write_append_file	Sink	2026-09-17T01:35:35Z	A	devlin	

ticker Functional and Technical Specification

This specification describes the behavior and current implementation of `/ticker`.

Contents

- ticker
- run.sh
- file_lib
- gen_text
- gen_uuid
- include_time_lib
- includes_lib
- proc_self_argv_lib
- read_time
- uuid_lib
- write_append_file
- README.md
- 96.png
- qjs.dll
- qjs.dylib
- qjs.so
- Shared Types
- Service Libraries

Processing Nodes

ticker

Specification

this workflow;

- reads the system time
- then packs it up with an uuid
- then adds a random word
- then finishes by adding the structure the structure to a file

Implementation Profile

`ticker` is a composite worker at `/ticker`. It is implemented in `c` for `c-native`.

Composition

The composite directly contains `run.sh` (processing unit), `Packet` (type), `file_lib` (service library), `gen_text` (transformer), `gen_uuid` (transformer), `include_time_lib` (service library), `includes_lib` (service library), `proc_self_argv_lib` (service library), `read_time` (generator), `uuid_lib` (service library), `write_append_file` (sink), `README.md` (processing unit), `96.png` (processing unit), `qjs.dll` (processing unit), `qjs.dylib` (processing unit), `qjs.so` (processing unit), `Signal` (type), `Packets` (type).

```
flowchart LR
  node_node_968f03ce_27b7_4776_a435_ae0d3a01085a["run.sh (processing unit)"]
  node_node_c4fb01af_8d9d_4a30_838f_8ab8f6328db8["Packet (type)"]
  node_node_b5041d57_c1e9_4f67_9e88_760c35c8efde["file_lib (service library)"]
  node_node_059814d5_2184_47b6_93b0_f754589cf34f["gen_text (transformer)"]
  node_node_627a3a3e_718f_4867_a71e_584b2d9b138b["gen_uuid (transformer)"]
  node_node_82203592_4fed_4fd4_b0bf_9b02a90b0ad0["include_time_lib (service library)"]
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b["includes_lib (service library)"]
  node_node_98b1de57_6b1a_4a96_8e55_6e9a11eb1fc4["proc_self_argv_lib (service library)"]
  node_node_c43e7a55_0e44_428b_972a_6dab999df396["read_time (generator)"]
  node_node_b153ded5_bb9d_438c_9854_84fa0e737261["uuid_lib (service library)"]
  node_node_3b600a6f_96a5_42b4_beae_b74338ccf170["write_append_file (sink)"]
  node_node_5b6c66ab_3d46_41fb_96ff_dee9c1738292["README.md (processing unit)"]
  node_node_4b032846_a445_4327_95db_4178aa447304["96.png (processing unit)"]
  node_node_078ec47e_e735_4cc2_a56a_34da3933ac8b["qjs.dll (processing unit)"]
  node_node_352d6efc_bce0_4273_8c4b_03c64e7764fc["qjs.dylib (processing unit)"]
  node_node_6a831982_71aa_47ad_ab14_fbfcbc9cd384["qjs.so (processing unit)"]
  node_node_34e10e8d_c693_4e56_91d9_f03334aafa9b["Signal (type)"]
  node_node_968adf46_fe80_45e7_a6c3_90d78a8697e4["Packets (type)"]
  node_node_b5041d57_c1e9_4f67_9e88_760c35c8efde -->|files| node_node_3b600a6f_96a5_42b4_beae_b74338ccf170
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b -->|includes5| node_node_b5041d57_c1e9_4f67_9e88_760c35c8efde
  node_node_059814d5_2184_47b6_93b0_f754589cf34f -->|ip_word| node_node_3b600a6f_96a5_42b4_beae_b74338ccf170
  node_node_627a3a3e_718f_4867_a71e_584b2d9b138b -->|ip_uuid| node_node_059814d5_2184_47b6_93b0_f754589cf34f
  node_node_c43e7a55_0e44_428b_972a_6dab999df396 -->|ip_time| node_node_627a3a3e_718f_4867_a71e_584b2d9b138b
  node_node_b153ded5_bb9d_438c_9854_84fa0e737261 -->|uuids| node_node_627a3a3e_718f_4867_a71e_584b2d9b138b
  node_node_82203592_4fed_4fd4_b0bf_9b02a90b0ad0 -->|tm| node_node_c43e7a55_0e44_428b_972a_6dab999df396
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b -->|includes3| node_node_82203592_4fed_4fd4_b0bf_9b02a90b0ad0
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b -->|includes2| node_node_98b1de57_6b1a_4a96_8e55_6e9a11eb1fc4
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b -->|includes6| node_node_b153ded5_bb9d_438c_9854_84fa0e737261
  node_node_98b1de57_6b1a_4a96_8e55_6e9a11eb1fc4 -->|argslib| node_node_3b600a6f_96a5_42b4_beae_b74338ccf170
```

run.sh

Current Implementation

```
tcc ticker.c -Wall -luuid -o ticker
./ticker sleep 1000000

# variant using the embedded tcc compiler
java -jar threadworks.jar tinycc --launcher java exe ticker.c -luuid -o ticker
javac -cp ../lib/tinycc-embed.jar TickerjLauncher.java
java -cp .:threadworks.jar TickerjLauncher --sleep 1000000
```

Implementation Profile

`run.sh` is a processing unit at `/ticker/run.sh`. It is implemented in `shellscript` for `file-export`.

file_lib

Specification

includes stdio.h or whatever is necessary for the source using the lib to have access to fopen write, fclose et al

Current Implementation

```
#ifndef DEF_THREADWORK_ERROR
#define THREADWORK_ERROR -1
#endif
```

Implementation Profile

`file_lib` is a service library at `/ticker/file_lib`. It is implemented in `c` for `c-native`; language and technology are inherited.

Interfaces and Dependencies

`file_lib` is provided access to `includes5:includes_lib`.

Dependency: includes5

`includes5` provides the service or library `includes_lib` as a processor-local dependency.

The provider is specified as follows:

#include declarations

gen_text

Specification

takes a Packet from ip_uuid and generates a random word string of length random_word_length using the dictionaries

consonants="bcdfgklmnprstuvxz" vowels="aeiou"

alternating consonants with vowels and starting with a consonant

finally it pushes the updated Packet to ip_word

Current Implementation

```
static const char consonants[] = "bcdfgklmnprstuvxz";
static const char vowels[] = "aeiou";

Packet item;
if (pop(ip_uuid, &item) != THREADWORK_OK) {
    return THREADWORK_OK;
}


// should raise a compilation error // azurty;

int length = (int)item.random_word_length;

if (length < 0) {
    item.isok = false;
    push(ip_word, &item);
    return THREADWORK_ERROR;
}

char *word = malloc((size_t)length + 1);

if (word == NULL) {
    item.isok = false;
    push(ip_word, &item);
    return THREADWORK_ERROR;
}

// should raise another compilation error // azurty;
for (int i = 0; i < length; ++i) {
    if ((i % 2) == 0) {
        word[i] = consonants[
            rand() % (sizeof(consonants) - 1)
        ];
    } else {
        word[i] = vowels[
            rand() % (sizeof(vowels) - 1)
        ];
    }
}

// shouldnt raise another compilation error because tcc is skibidy // azurty;
word[length] = '\0';

item.random_word = word;
printf("gen_text.timestamp:%s\n",item.timestamp != NULL ? item.timestamp : "null");
printf("gen_text.uuid:%s\n",item.uuid != NULL ? item.uuid : "null");
printf("gen_text.random_word:%s\n",item.random_word != NULL ? item.random_word : "null");

push(ip_word, &item);
```

Implementation Profile

`gen_text` is a transformer at `/ticker/gen_text`. It is implemented in `c` for `c-native`; language and technology are inherited.

Interfaces and Dependencies

`gen_text` receives the following inputs: `ip_uuid`; produces the following outputs: `ip_word`.

Input: ip_uuid

`ip_uuid` receives information packets of type `Packet` from `gen_uuid` through `gen_uuid.out -> gen_text.in`.

`Packet` has the following fields: `timestamp`: `string`, `uuid`: `string`, `random_word`: `string`, `random_word_length`: `number`, `isok`: `boolean`.

Output: ip_word

`ip_word` publishes information packets of type `Packet` to `write_append_file` through `gen_text.out -> write_append_file.in`.

`Packet` has the following fields: `timestamp`: `string`, `uuid`: `string`, `random_word`: `string`, `random_word_length`: `number`, `isok`: `boolean`.

gen_uuid

Specification

takes a Packet from ip_time, fills in its uuid field, sets random_word_length to a number between 5 and 25, and pushes the updated packet to ip_uuid

Current Implementation

```
Packet item;
if (pop(ip_time, &item) != THREADWORK_OK) {
    return THREADWORK_OK;
}

// should raise a compilation error // -gibberish--

uuid_t uuid_value;
char *uuid_string = malloc(37);

if (uuid_string == NULL) {
    item.isok = false;
    push(ip_uuid, &item);
    return THREADWORK_ERROR;
}

uuids__uuid_generate_random(uuid_value);
uuids__uuid_unparse_lower(uuid_value, uuid_string);

item.uuid = uuid_string;
item.random_word_length = 5 + (uuid_value[0] % 21);
printf("gen_text.timestamp:%s\n",item.timestamp != NULL ? item.timestamp : "null");
printf("gen_text.uuid:%s\n",item.uuid != NULL ? item.uuid : "null");

push(ip_uuid, &item);

pu
```

Implementation Profile

`gen_uuid` is a transformer at `/ticker/gen_uuid`. It is implemented in `c` for `c-native`; language and technology are inherited.

Interfaces and Dependencies

`gen_uuid` receives the following inputs: `ip_time`; is provided access to `uuids:uuid_lib`; produces the following outputs: `ip_uuid`.

Input: ip_time

`ip_time` receives information packets of type `Packet` from `read_time` through `read_time.out -> gen_uuid.in`.

`Packet` has the following fields: `timestamp`: `string`, `uuid`: `string`, `random_word`: `string`, `random_word_length`: `number`, `isok`: `boolean`.

Output: ip_uuid

`ip_uuid` publishes information packets of type `Packet` to `gen_text` through `gen_uuid.out -> gen_text.in`.

`Packet` has the following fields: `timestamp`: `string`, `uuid`: `string`, `random_word`: `string`, `random_word_length`: `number`, `isok`: `boolean`.

Dependency: uuids

`uuids` provides the service or library `uuid_lib` as a processor-local dependency.

The provider is specified as follows:

includes the necessary headers to generate uuids is gona be used by various functions

include_time_lib

Specification

creates a function that gets the system time and returns it as an int

Current Implementation

```
#ifndef TIME_LIB_H
#define TIME_LIB_H

#include <stdbool.h>
#include <stdio.h>
#include <time.h>

#endif /* TIME_LIB_H */
```

Implementation Profile

`include_time_lib` is a service library at `/ticker/include_time_lib`. It is implemented in `c` for `c-native`; language and technology are inherited.

Interfaces and Dependencies

`include_time_lib` is provided access to `includes3:includes_lib`.

Dependency: includes3

`includes3` provides the service or library `includes_lib` as a processor-local dependency.

The provider is specified as follows:

#include declarations

includes_lib

Specification

#include declarations

Current Implementation

```
#ifndef INCLUDES_LIB_H
#define INCLUDES_LIB_H

#include <stdbool.h>
#include <stdio.h>
#include <time.h>
#include <string.h>
#include <stdbool.h>
#include <uuid/uuid.h>
#include <stdio.h>
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

#endif /* INCLUDES_LIB_H */
```

Implementation Profile

`includes_lib` is a service library at `/ticker/includes_lib`. It is implemented in `c` for `c-native`; language and technology are inherited.

proc_self_argv_lib

Specification

reads the commandline from "/proc/self/cmdline"

Current Implementation

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define ARGPARSE_NOTERM  -1
#define ARGPARSE_NOVALUE -2
#define ARGPARSE_OOBOUND -3

static char  *argparse_buffer = NULL;
static char **argparse_argv   = NULL;
static int    argparse_argc   = 0;
static int    argparse_loaded = 0;


/* ------------------------------------------------------- */
/* internal loading                                        */
/* ------------------------------------------------------- */

static int argparse_load(void)
{

    if (argparse_loaded)
        return 0;

    FILE *f = fopen("/proc/self/cmdline", "rb");
    if (f == NULL)
        return ARGPARSE_NOTERM;

    size_t capacity = 256;
```

```
    size_t size = 0;

    argparse_buffer = malloc(capacity);
    if (argparse_buffer == NULL) {
        fclose(f);
        return ARGPARSE_OOBOUND;
    }

    for (;;) {
        if (size == capacity) {
            capacity *= 2;

            char *new_buffer = realloc(argparse_buffer, capacity);
            if (new_buffer == NULL) {
                free(argparse_buffer);
                argparse_buffer = NULL;
                fclose(f);
                return ARGPARSE_OOBOUND;
            }

            argparse_buffer = new_buffer;
        }

        size_t n = fread(
            argparse_buffer + size,
            1,
            capacity - size,
            f
        );

        size += n;

        if (n == 0)
            break;
    }

    fclose(f);

    if (size == 0)
        return ARGPARSE_NOTERM;

    /*
     * /proc/self/cmdline normally ends in \0,
     * but guarantee it ourselves.
     */
    if (argparse_buffer[size - 1] != '\0') {
        if (size == capacity) {
            capacity++;

            char *new_buffer = realloc(argparse_buffer, capacity);
            if (new_buffer == NULL)
                return ARGPARSE_OOBOUND;

            argparse_buffer = new_buffer;
        }

        argparse_buffer[size++] = '\0';
    }

    /* count arguments */
    for (size_t i = 0; i < size; i++) {
        if (argparse_buffer[i] == '\0')
            argparse_argc++;
    }

    argparse_argv = malloc(
        sizeof(char *) * (size_t)argparse_argc
    );

    if (argparse_argv == NULL)
        return ARGPARSE_OOBOUND;

    int index = 0;
    char *p = argparse_buffer;
    char *end = argparse_buffer + size;

    while (p < end && index < argparse_argc) {
        argparse_argv[index++] = p;
        p += strlen(p) + 1;
    }

    argparse_loaded = 1;
    return 0;
}


/* ------------------------------------------------------- */
/* helpers                                                 */
/* ------------------------------------------------------- */

static int option_like(const char *s)
{
    return s != NULL &&
           s[0] == '-' &&
           s[1] != '\0';
}
```

```
/* ------------------------------------------------------- */
/* --name value                                            */
/* ------------------------------------------------------- */

int option(const char *name, char **option_value)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (name == NULL || option_value == NULL)
        return ARGPARSE_NOTERM;

    *option_value = NULL;

    for (int i = 1; i < argparse_argc; i++) {

        const char *arg = argparse_argv[i];

        if (strncmp(arg, "--", 2) != 0)
            continue;

        const char *actual_name = arg + 2;

        if (strcmp(actual_name, name) == 0) {

            if (i + 1 >= argparse_argc)
                return ARGPARSE_NOVALUE;

            if (option_like(argparse_argv[i + 1]))
                return ARGPARSE_NOVALUE;

            *option_value = argparse_argv[i + 1];
            return i;
        }
    }

    return ARGPARSE_NOTERM;
}


/* ------------------------------------------------------- */
/* -flag                                                   */
/* ------------------------------------------------------- */

int flag(const char *name)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (name == NULL)
        return ARGPARSE_NOTERM;

    for (int i = 1; i < argparse_argc; i++) {

        const char *arg = argparse_argv[i];

        if (arg[0] != '-' || arg[1] == '-')
            continue;

        if (strcmp(arg + 1, name) == 0)
            return i;
    }

    return ARGPARSE_NOTERM;
}


/* ------------------------------------------------------- */
/* -p value                                                */
/* ------------------------------------------------------- */

int short_option(const char *name, char **option_value)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (name == NULL || option_value == NULL)
        return ARGPARSE_NOTERM;

    *option_value = NULL;

    for (int i = 1; i < argparse_argc; i++) {

        const char *arg = argparse_argv[i];

        if (arg[0] != '-' || arg[1] == '-')
            continue;

        if (strcmp(arg + 1, name) == 0) {

            if (i + 1 >= argparse_argc)
                return ARGPARSE_NOVALUE;

            if (option_like(argparse_argv[i + 1]))
                return ARGPARSE_NOVALUE;
```

```
            *option_value = argparse_argv[i + 1];
            return i;
        }
    }

    return ARGPARSE_NOTERM;
}


/* ------------------------------------------------------- */
/* literal word                                            */
/* ------------------------------------------------------- */

int word(const char *name)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (name == NULL)
        return ARGPARSE_NOTERM;

    for (int i = 1; i < argparse_argc; i++) {

        if (strcmp(argparse_argv[i], name) == 0)
            return i;
    }

    return ARGPARSE_NOTERM;
}


/* ------------------------------------------------------- */
/* everything from argv[startpos] onward                   */
/* ------------------------------------------------------- */

int varargs(int startpos, char **results, int capacity)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (startpos < 0 || startpos >= argparse_argc)
        return ARGPARSE_OOBOUND;

    int count = argparse_argc - startpos;

    /*
     * +1 because results is NULL terminated.
     */
    if (results == NULL || capacity < count + 1)
        return ARGPARSE_OOBOUND;

    for (int i = 0; i < count; i++)
        results[i] = argparse_argv[startpos + i];

    results[count] = NULL;

    return startpos;
}


/* ------------------------------------------------------- */
/* raw argv access                                         */
/* ------------------------------------------------------- */

int get_arg(int n, char **name, char **value)
{
    if (argparse_load() < 0)
        return ARGPARSE_NOTERM;

    if (n < 0 || n >= argparse_argc)
        return ARGPARSE_OOBOUND;

    if (name == NULL || value == NULL)
        return ARGPARSE_OOBOUND;

    char *arg = argparse_argv[n];

    *value = NULL;

    /*
     * --option
     */
    if (strncmp(arg, "--", 2) == 0) {

        *name = arg + 2;

        if (n + 1 >= argparse_argc ||
            option_like(argparse_argv[n + 1]))
            return ARGPARSE_NOVALUE;

        *value = argparse_argv[n + 1];
        return n;
    }

    /*
     * -flag / -option
     */
```

```
    if (arg[0] == '-' && arg[1] != '\0') {

        *name = arg + 1;

        /*
         * Could be a flag, therefore absence of a following
         * value is not considered an error here.
         */
        if (n + 1 < argparse_argc &&
            !option_like(argparse_argv[n + 1])) {
            *value = argparse_argv[n + 1];
        }

        return n;
    }

    /*
     * ordinary word
     */
    *name = arg;
    return n;
}
```

Implementation Profile

`proc_self_argv_lib` is a service library at `/ticker/proc_self_argv_lib`. It is implemented in `c` for `c-native`; language and technology are inherited.

Interfaces and Dependencies

`proc_self_argv_lib` is provided access to `includes2:includes_lib`.

Dependency: includes2

`includes2` provides the service or library `includes_lib` as a processor-local dependency.

The provider is specified as follows:

#include declarations

read_time

Specification

reads system time and puts it in a Packet and then pushes it into the ip_time output

Current Implementation

```
time_t now = time(NULL);
struct tm time_info;
char timestamp[32];

// should raise a compilation error // some_nonexisten_function==!!


Packet packet = {
    .timestamp = timestamp,
    .uuid = NULL,
    .random_word = NULL,
    .random_word_length = 0
};

if (now != (time_t)-1 &&
    localtime_r(&now, &time_info) != NULL &&
    tm__strftime(
        timestamp,
        sizeof(timestamp),
        "%Y-%m-%dT%H:%M:%S%z",
        &time_info
    ) > 0) {
    packet.isok = true;
} else {
    timestamp[0] = '\0';
}


packet.timestamp = strdup(timestamp);

printf("read_time.timestamp:%s\n",packet.timestamp != NULL ? packet.timestamp : "null");

push(ip_time, &packet);
```

Implementation Profile

`read_time` is a generator at `/ticker/read_time`. It is implemented in `c` for `c-native`; language and technology are inherited.

Interfaces and Dependencies

`read_time` is provided access to `tm:include_time_lib`; produces the following outputs: `ip_time`.

Output: ip_time

`ip_time` publishes information packets of type `Packet` to `gen_uuid` through `read_time.out -> gen_uuid.in`.

`Packet` has the following fields: `timestamp`: `string`, `uuid`: `string`, `random_word`: `string`, `random_word_length`: `number`, `isok`: `boolean`.

Dependency: tm

`tm` provides the service or library `include_time_lib` as a processor-local dependency.

The provider is specified as follows:

creates a function that gets the system time and returns it as an int

uuid_lib

Specification

includes the necessary headers to generate uuids is gona be used by various functions

Current Implementation

```
#ifndef UUID_LIB_H
#define UUID_LIB_H

#include <uuid/uuid.h>
#include <stdbool.h>
#endif /* UUID_LIB_H */
```

Implementation Profile

`uuid_lib` is a service library at `/ticker/uuid_lib`. It is implemented in `c` for `c-native`; language and technology are inherited.

Interfaces and Dependencies

`uuid_lib` is provided access to `includes6:includes_lib`.

Dependency: includes6

`includes6` provides the service or library `includes_lib` as a processor-local dependency.

The provider is specified as follows:

#include declarations

write_append_file

Specification

takes a Packet from ip_word, serializes it as a CSV record, and appends the result to "ticker_out.csv"

Current Implementation

```
Packet item;

char *value;
int sleepval=5000;
int p = argslib__option("sleep", &value);
if(p>=0){
    sleepval=atoi(value);
}
printf("write_append_file.sleepvalue:%d\n",sleepval);


if (pop(ip_word, &item) != THREADWORK_OK) {
    return THREADWORK_OK;
}

FILE *file = fopen("ticker_out.csv", "a");
if (file == NULL) {
    return THREADWORK_ERROR;
}

int fprintf_result = fprintf(
    file,
    "\"%s\",\"%s\",\"%s\",%.0f,%s\n",
    item.timestamp != NULL ? item.timestamp : "",
    item.uuid != NULL ? item.uuid : "",
    item.random_word != NULL ? item.random_word : "",
    item.random_word_length,
    item.isok ? "true" : "false"
);
printf("write_append_file.timestamp:%s\n",item.timestamp != NULL ? item.timestamp : "null");
printf("write_append_file.uuid:%s\n",item.uuid != NULL ? item.uuid : "null");
printf("write_append_file.random_word:%s\n",item.random_word != NULL ? item.random_word : "null");

if (fprintf_result < 0) {
    fclose(file);
    return THREADWORK_ERROR;
}

if (fclose(file) != 0) {
    return THREADWORK_ERROR;
}

free(item.timestamp);

printf("write_append_file.sleeping 5000 ns \n");
usleep(sleepval);
```

Implementation Profile

`write_append_file` is a sink at `/ticker/write_append_file`. It is implemented in `c` for `c-native`; language and technology are inherited.

Interfaces and Dependencies

`write_append_file` receives the following inputs: `ip_word`; is provided access to `argslib:proc_self_argv_lib`, `files:file_lib`.

Input: ip_word

`ip_word` receives information packets of type `Packet` from `gen_text` through `gen_text.out -> write_append_file.in`.

`Packet` has the following fields: `timestamp`: `string`, `uuid`: `string`, `random_word`: `string`, `random_word_length`: `number`, `isok`: `boolean`.

Dependency: argslib

`argslib` provides the service or library `proc_self_argv_lib` as a processor-local dependency.

The provider is specified as follows:

reads the commandline from "/proc/self/cmdline"

Dependency: files

`files` provides the service or library `file_lib` as a processor-local dependency.

The provider is specified as follows:

includes stdio.h or whatever is necessary for the source using the lib to have access to fopen write, fclose et al

README.md

Current Implementation

```
this is something that i wrote
```

Implementation Profile

`README.md` is a processing unit at `/ticker/README.md`. It is implemented in `shellscript` for `file-export`.

96.png

Implementation Profile

`96.png` is a processing unit at `/ticker/96.png`. It is implemented in `plain` for `file-export`; language is inherited.

qjs.dll

Implementation Profile

`qjs.dll` is a processing unit at `/ticker/qjs.dll`. It is implemented in `plain` for `file-export`; language is inherited.

qjs.dylib

Implementation Profile

`qjs.dylib` is a processing unit at `/ticker/qjs.dylib`. It is implemented in `plain` for `file-export`; language is inherited.

qjs.so

Implementation Profile

`qjs.so` is a processing unit at `/ticker/qjs.so`. It is implemented in `plain` for `file-export`; language is inherited.

Type and Service Annexes

Shared Types

- `Packet`
- `Packets`
- `Signal`

Service Libraries

- `file_lib` at `/ticker/file_lib`
- `include_time_lib` at `/ticker/include_time_lib`
- `includes_lib` at `/ticker/includes_lib`
- `proc_self_argv_lib` at `/ticker/proc_self_argv_lib`
- `uuid_lib` at `/ticker/uuid_lib`

ticker Annexes ; Components Breakdown

Generated from Threadwork component specifications for `/ticker`.

Contents

- ticker
- run.sh
- file_lib
- gen_text
- gen_uuid
- include_time_lib
- includes_lib
- proc_self_argv_lib
- read_time
- uuid_lib
- write_append_file
- README.md
- 96.png
- qjs.dll
- qjs.dylib
- qjs.so

ticker

- **Path:** `/ticker`
- **Role:** `CompositeWorker`
- **Language:** `c`
- **Technology:** `c-native`

Direct Children

Name	Stereotype
96.png	ProcessingUnit
file_lib	ServiceLibrary
gen_text	Transformer
gen_uuid	Transformer
include_time_lib	ServiceLibrary
includes_lib	ServiceLibrary
Packet	Type
Packets	Type
proc_self_argv_lib	ServiceLibrary
qjs.dll	ProcessingUnit
qjs.dylib	ProcessingUnit
qjs.so	ProcessingUnit
read_time	Generator
README.md	ProcessingUnit
run.sh	ProcessingUnit
Signal	Type
uuid_lib	ServiceLibrary
write_append_file	Sink

Flow Diagram (Mermaid)

```
flowchart LR
  node_node_4b032846_a445_4327_95db_4178aa447304["96.png (ProcessingUnit)"]
  node_node_b5041d57_c1e9_4f67_9e88_760c35c8efde["file_lib (ServiceLibrary)"]
  node_node_059814d5_2184_47b6_93b0_f754589cf34f["gen_text (Transformer)"]
  node_node_627a3a3e_718f_4867_a71e_584b2d9b138b["gen_uuid (Transformer)"]
  node_node_82203592_4fed_4fd4_b0bf_9b02a90b0ad0["include_time_lib (ServiceLibrary)"]
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b["includes_lib (ServiceLibrary)"]
  node_node_c4fb01af_8d9d_4a30_838f_8ab8f6328db8["Packet (Type)"]
  node_node_968adf46_fe80_45e7_a6c3_90d78a8697e4["Packets (Type)"]
  node_node_98b1de57_6b1a_4a96_8e55_6e9a11eb1fc4["proc_self_argv_lib (ServiceLibrary)"]
  node_node_078ec47e_e735_4cc2_a56a_34da3933ac8b["qjs.dll (ProcessingUnit)"]
  node_node_352d6efc_bce0_4273_8c4b_03c64e7764fc["qjs.dylib (ProcessingUnit)"]
  node_node_6a831982_71aa_47ad_ab14_fbfcbc9cd384["qjs.so (ProcessingUnit)"]
  node_node_c43e7a55_0e44_428b_972a_6dab999df396["read_time (Generator)"]
  node_node_5b6c66ab_3d46_41fb_96ff_dee9c1738292["README.md (ProcessingUnit)"]
  node_node_968f03ce_27b7_4776_a435_ae0d3a01085a["run.sh (ProcessingUnit)"]
  node_node_34e10e8d_c693_4e56_91d9_f03334aafa9b["Signal (Type)"]
  node_node_b153ded5_bb9d_438c_9854_84fa0e737261["uuid_lib (ServiceLibrary)"]
  node_node_3b600a6f_96a5_42b4_beae_b74338ccf170["write_append_file (Sink)"]
  node_node_98b1de57_6b1a_4a96_8e55_6e9a11eb1fc4 -->|argslib| node_node_3b600a6f_96a5_42b4_beae_b74338ccf170
  node_node_b5041d57_c1e9_4f67_9e88_760c35c8efde -->|files| node_node_3b600a6f_96a5_42b4_beae_b74338ccf170
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b -->|includes2| node_node_98b1de57_6b1a_4a96_8e55_6e9a11eb1fc4
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b -->|includes3| node_node_82203592_4fed_4fd4_b0bf_9b02a90b0ad0
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b -->|includes5| node_node_b5041d57_c1e9_4f67_9e88_760c35c8efde
  node_node_b886239f_e4d3_490b_be5d_4e0d0792565b -->|includes6| node_node_b153ded5_bb9d_438c_9854_84fa0e737261
  node_node_c43e7a55_0e44_428b_972a_6dab999df396 -->|ip_time| node_node_627a3a3e_718f_4867_a71e_584b2d9b138b
  node_node_627a3a3e_718f_4867_a71e_584b2d9b138b -->|ip_uuid| node_node_059814d5_2184_47b6_93b0_f754589cf34f
  node_node_059814d5_2184_47b6_93b0_f754589cf34f -->|ip_word| node_node_3b600a6f_96a5_42b4_beae_b74338ccf170
  node_node_82203592_4fed_4fd4_b0bf_9b02a90b0ad0 -->|tm| node_node_c43e7a55_0e44_428b_972a_6dab999df396
  node_node_b153ded5_bb9d_438c_9854_84fa0e737261 -->|uuids| node_node_627a3a3e_718f_4867_a71e_584b2d9b138b
```

Specification

this workflow;

- reads the system time
- then packs it up with an uuid
- then adds a random word
- then finishes by adding the structure the structure to a file

run.sh

- **Path:** `/ticker/run.sh`
- **Role:** `ProcessingUnit`
- **Language:** `shellscript`
- **Technology:** `file-export`

file_lib

- **Path:** `/ticker/file_lib`
- **Role:** `ServiceLibrary`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Specification

includes stdio.h or whatever is necessary for the source using the lib to have access to fopen write, fclose et al

Used Libraries and Capabilities

includes5

- **Library:** `includes_lib`
- **Library path:** `/ticker/includes_lib`
- **Classification:** `DependencyInjection`
- **Interaction:** `lib`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Library specification

#include declarations

gen_text

- **Path:** `/ticker/gen_text`
- **Role:** `Transformer`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Specification

takes a Packet from ip_uuid and generates a random word string of length random_word_length using the dictionaries

consonants="bcdfgklmnprstuvxz" vowels="aeiou"

alternating consonants with vowels and starting with a consonant

finally it pushes the updated Packet to ip_word

Inputs

Name	Type	Source	Port
ip_uuid	Packet	gen_uuid	in

Incoming Link Contracts

ip_uuid:Packet

- **Link ID:** `link_8e881876-8d2a-41a8-92a4-2ee3f2c839ce`
- **From:** `gen_uuid.out`
- **To:** `gen_text.in`
- **Transport:** `in-process`
- **Interaction:** `data`
- **Classification:** `Transport`

Shared type: `Packet`

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Outputs

Name	Type	Target	Port
ip_word	Packet	write_append_file	out

Outgoing Link Contracts

ip_word:Packet

- **Link ID:** `link_dd5fcd39-8d54-4310-b6fe-242d927f0ec5`
- **From:** `gen_text.out`
- **To:** `write_append_file.in`
- **Transport:** `in-process`
- **Interaction:** `data`
- **Classification:** `Transport`

Shared type: `Packet`

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Used Types

Packet

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

gen_uuid

- **Path:** `/ticker/gen_uuid`
- **Role:** `Transformer`
- **Language:** Inferred (c)
- **Technology:** Inferred (`c-native`)

Specification

takes a Packet from ip_time, fills in its uuid field, sets random_word_length to a number between 5 and 25, and pushes the updated packet to ip_uuid

Inputs

Name	Type	Source	Port
ip_time	Packet	read_time	in

Incoming Link Contracts

ip_time:Packet

- **Link ID:** `link_d1b9d286-d43a-439a-bee0-e66bf2943400`
- **From:** `read_time.out`
- **To:** `gen_uuid.in`
- **Transport:** `in-process`
- **Interaction:** `data`
- **Classification:** `Transport`

Shared type: `Packet`

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Outputs

Name	Type	Target	Port
ip_uuid	Packet	gen_text	out

Outgoing Link Contracts

ip_uuid:Packet

- **Link ID:** `link_8e881876-8d2a-41a8-92a4-2ee3f2c839ce`
- **From:** `gen_uuid.out`
- **To:** `gen_text.in`
- **Transport:** `in-process`
- **Interaction:** `data`
- **Classification:** `Transport`

Shared type: `Packet`

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Used Types

Packet

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Used Libraries and Capabilities

uuids

- **Library:** `uuid_lib`
- **Library path:** `/ticker/uuid_lib`
- **Classification:** `DependencyInjection`
- **Interaction:** `lib`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Library specification

includes the necessary headers to generate uuids is gona be used by various functions

include_time_lib

- **Path:** `/ticker/include_time_lib`
- **Role:** `ServiceLibrary`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Specification

creates a function that gets the system time and returns it as an int

Used Libraries and Capabilities

includes3

- **Library:** `includes_lib`
- **Library path:** `/ticker/includes_lib`
- **Classification:** `DependencyInjection`
- **Interaction:** `lib`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Library specification

#include declarations

includes_lib

- **Path:** `/ticker/includes_lib`
- **Role:** `ServiceLibrary`
- **Language:** Inferred (c)
- **Technology:** Inferred (`c-native`)

Specification

#include declarations

proc_self_argv_lib

- **Path:** `/ticker/proc_self_argv_lib`
- **Role:** `ServiceLibrary`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Specification

reads the commandline from "/proc/self/cmdline"

Used Libraries and Capabilities

includes2

- **Library:** `includes_lib`
- **Library path:** `/ticker/includes_lib`
- **Classification:** `DependencyInjection`
- **Interaction:** `lib`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Library specification

#include declarations

read_time

- **Path:** `/ticker/read_time`
- **Role:** `Generator`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Specification

reads system time and puts it in a Packet and then pushes it into the ip_time output

Outputs

Name	Type	Target	Port
ip_time	Packet	gen_uuid	out

Outgoing Link Contracts

ip_time:Packet

- **Link ID:** `link_d1b9d286-d43a-439a-bee0-e66bf2943400`
- **From:** `read_time.out`
- **To:** `gen_uuid.in`
- **Transport:** `in-process`
- **Interaction:** `data`
- **Classification:** `Transport`

Shared type: `Packet`

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Used Types

Packet

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Used Libraries and Capabilities

tm

- **Library:** `include_time_lib`
- **Library path:** `/ticker/include_time_lib`
- **Classification:** `DependencyInjection`
- **Interaction:** `lib`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Library specification

creates a function that gets the system time and returns it as an int

uuid_lib

- **Path:** `/ticker/uuid_lib`
- **Role:** `ServiceLibrary`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Specification

includes the necessary headers to generate uuids is gona be used by various functions

Used Libraries and Capabilities

includes6

- **Library:** `includes_lib`
- **Library path:** `/ticker/includes_lib`
- **Classification:** `DependencyInjection`
- **Interaction:** `lib`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Library specification

#include declarations

write_append_file

- **Path:** `/ticker/write_append_file`
- **Role:** `Sink`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Specification

takes a Packet from ip_word, serializes it as a CSV record, and appends the result to "ticker_out.csv"

Inputs

Name	Type	Source	Port
ip_word	Packet	gen_text	in

Incoming Link Contracts

ip_word:Packet

- **Link ID:** `link_dd5fcd39-8d54-4310-b6fe-242d927f0ec5`
- **From:** `gen_text.out`
- **To:** `write_append_file.in`
- **Transport:** `in-process`
- **Interaction:** `data`
- **Classification:** `Transport`

Shared type: `Packet`

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Used Types

Packet

Field	Type	Reference
timestamp	string	No
uuid	string	No
random_word	string	No
random_word_length	number	No
isok	boolean	No

Used Libraries and Capabilities

argslib

- **Library:** `proc_self_argv_lib`
- **Library path:** `/ticker/proc_self_argv_lib`
- **Classification:** `DependencyInjection`
- **Interaction:** `lib`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Library specification

reads the commandline from "/proc/self/cmdline"

files

- **Library:** `file_lib`
- **Library path:** `/ticker/file_lib`
- **Classification:** `DependencyInjection`
- **Interaction:** `lib`
- **Language:** Inferred (`c`)
- **Technology:** Inferred (`c-native`)

Library specification

includes stdio.h or whatever is necessary for the source using the lib to have access to fopen write, fclose et al

README.md

- **Path:** `/ticker/README.md`
- **Role:** `ProcessingUnit`
- **Language:** `shellscript`
- **Technology:** `file-export`

96.png

- **Path:** `/ticker/96.png`
- **Role:** `ProcessingUnit`
- **Language:** Inferred (`plain`)
- **Technology:** `file-export`

qjs.dll

- **Path:** `/ticker/qjs.dll`
- **Role:** `ProcessingUnit`
- **Language:** Inferred (`plain`)
- **Technology:** `file-export`

qjs.dylib

- **Path:** `/ticker/qjs.dylib`
- **Role:** `ProcessingUnit`
- **Language:** Inferred (`plain`)
- **Technology:** `file-export`

qjs.so

- **Path:** `/ticker/qjs.so`
- **Role:** `ProcessingUnit`
- **Language:** Inferred (`plain`)
- **Technology:** `file-export`